

Discovering Latent Structure in U.S. Retail Electricity Markets: A Comparative Study of K-Means, DBSCAN, and Hierarchical Clustering

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Abstract—This work applies unsupervised clustering to monthly retail electricity sales data from the U.S. Energy Information Administration (EIA), covering 62 states and territories from January 2001 to January 2026. The unit of analysis is a state-month observation described by eight engineered features capturing trend, seasonality, volatility, and price dynamics. Three algorithms—K-Means, DBSCAN, and Agglomerative Hierarchical Clustering—are applied to the same preprocessed feature matrix and evaluated using three internal validation metrics: Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index. A hyperparameter sensitivity study is performed for each algorithm, and a feature ablation experiment compares results across the full feature set versus two domain-motivated subsets (temporal and economic). All three algorithms converge on a bimodal partition of the data, yielding a Silhouette Score of approximately 0.72, which reveals a robust structural divide between high-volume, standard-price markets and a small cluster of extreme-volatility observations. DBSCAN additionally identifies 77 anomalous state-month records (0.4% of the dataset) that the partition-based methods absorb into clusters, offering complementary domain insight into consumption outliers.

Index Terms—Unsupervised Learning, Clustering, K-Means, DBSCAN, Hierarchical Clustering, Electricity Consumption, Energy Data, Silhouette Score, Feature Ablation

I. INTRODUCTION

Understanding the latent structure of energy consumption is essential for policy design, grid planning, and market regulation. Unlike supervised approaches that require labeled outcome data, clustering methods discover natural groupings from observational records alone, making them suitable for exploratory analysis of large administrative datasets [1].

The U.S. retail electricity market generates a rich longitudinal record through the Energy Information Administration (EIA) monthly survey, covering all states and major sectors. Prior work in Challenge 2 addressed this domain using supervised and semi-supervised classification to predict whether a given state-month observation falls within the top consumption decile. The present study removes the label entirely and asks a more fundamental question: *what groupings emerge from the data when no outcome signal is imposed?*

Three complementary algorithms are compared. K-Means [2], [3] assumes convex, roughly equal-size clusters and serves as the interpretable baseline. DBSCAN [4] discovers clusters of arbitrary shape and explicitly marks low-density points as

noise, making it suitable for detecting anomalous consumption regimes. Agglomerative Hierarchical Clustering [5] builds a dendrogram that reveals structure at multiple scales, providing a cross-check on the flat partitions produced by the other two methods.

The main contributions of this paper are:

- (i) A rigorous comparative study of three clustering algorithms on the same preprocessed EIA feature matrix;
- (ii) A hyperparameter sensitivity analysis with documented selection criteria;
- (iii) A feature ablation experiment quantifying the contribution of temporal versus economic features; and
- (iv) A domain interpretation of the discovered clusters in terms of U.S. electricity market structure.

II. RELATED WORK

Clustering has been widely applied to electricity load data. Chicco [6] surveys load-profile clustering methods for demand-side management, showing that K-Means and hierarchical methods dominate practical deployments due to their interpretability. Räsänen and Kolehmainen [7] demonstrate that feature engineering—particularly rolling statistics and seasonal encodings—substantially improves cluster quality for energy time series. More recently, density-based methods have gained traction for anomaly detection in smart-meter data [8], a use case aligned with DBSCAN’s noise identification capability.

The internal validation metrics used here (Silhouette, Davies–Bouldin, Calinski–Harabasz) follow the recommendations of Arbelaitz et al. [9], who conduct the most comprehensive empirical comparison of cluster validity indices to date.

III. DATA AND PREPROCESSING

A. Dataset

The dataset is obtained from the EIA Retail Sales API, filtering for the all-sectors aggregate (`sectorid = ALL`) to ensure comparability across states. After filtering, the raw dataset contains 18,662 state-month records spanning 62 states and territories from January 2001 to January 2026. Following feature engineering and removal of rows with insufficient history for lag computation, the final matrix contains **17,918 observations** and **8 features**.

B. Feature Engineering

All features are constructed using only past observations to avoid look-ahead bias. The full feature set is described in Table I.

TABLE I
ENGINEERED FEATURES USED FOR CLUSTERING

Feature	Description
roll12_mean	12-month trailing mean of sales (MWh)
roll12_std	12-month trailing standard deviation (volatility proxy)
yoy_growth	Year-over-year percentage change, clipped to $[-5, 5]$
price	Retail electricity price (¢/kWh)
revenue_per_mwh	Revenue divided by sales (normalised price proxy)
month_sin	$\sin(2\pi \cdot \text{month}/12)$ (Fourier seasonality)
month_cos	$\cos(2\pi \cdot \text{month}/12)$ (Fourier seasonality)
year_trend	Linear month index from January 2001

C. Preprocessing Pipeline

The preprocessing pipeline is applied identically to all three algorithms to ensure fair comparison. Features are standardised using `StandardScaler` (zero mean, unit variance). Principal Component Analysis (PCA) is then applied, retaining components that explain at least 90% of the total variance; this yields **5 components** accounting for 90.8% of variance. The PCA-reduced matrix is the input to all clustering algorithms.

IV. METHODOLOGY

A. K-Means

K-Means partitions the data into k non-overlapping clusters by minimising within-cluster sum of squared distances [3]. A sweep over $k \in \{2, \dots, 12\}$ is performed with `init=k-means++` and `n_init=10` to mitigate local-optima sensitivity. The optimal k is selected by the Silhouette Score; inertia is reported for the elbow plot but is not used as the sole criterion. Stability is assessed by repeating the best configuration with seeds $\{42, 123, 777\}$.

B. DBSCAN

DBSCAN [4] discovers clusters of arbitrary shape without requiring k to be specified, and marks low-density points as noise (label -1). The neighbourhood radius ε is estimated from the k -nearest-neighbour distance plot: the 90th percentile of the 5-NN distance distribution is used as the elbow heuristic, yielding $\varepsilon_{\text{elbow}} = 0.547$. A grid search over $\varepsilon \in [0.273, 1.093]$ (8 values) and `min_samples` $\in \{5, 10\}$ is conducted.

To avoid degenerate solutions, configurations with more than 30 clusters or a noise fraction exceeding 40% are excluded. The best configuration is selected by a composite score:

$$S_{\text{composite}} = \text{Sil} - 0.3 \cdot \text{DB} - 0.5 \cdot f_{\text{noise}} \quad (1)$$

where `Sil` is the Silhouette Score, `DB` is the Davies–Bouldin Index, and f_{noise} is the noise fraction. This formulation balances cluster compactness, separation, and coverage.

C. Hierarchical Clustering

Agglomerative Clustering [5] is applied with linkage criteria `ward`, `complete`, and `average`. The dendrogram is computed on a stratified sample of 2,000 observations. Unlike the original implementation that fixed k to the K-Means result, the corrected version independently sweeps $k \in \{2, 3, 4\}$ for each linkage criterion, selecting the best $(k, \text{linkage})$ pair by Silhouette Score.

V. EXPERIMENTAL RESULTS

A. PCA Dimensionality Reduction

Prior to clustering, PCA is applied to the standardised 8-dimensional feature matrix to reduce noise and computational cost while preserving the dominant variance structure. Fig. 1 plots the cumulative explained variance as a function of the number of retained components. The curve rises steeply for the first two components—which together account for 53.5% of total variance—and then grows more gradually, crossing the 90% threshold at exactly $n = 5$ components (90.8% retained). Beyond $n = 6$ the marginal gain per additional component falls below 1%, confirming that five components capture the essential structure of the data without retaining noise dimensions. The 5-component PCA representation is therefore used as the common input to all three clustering algorithms.

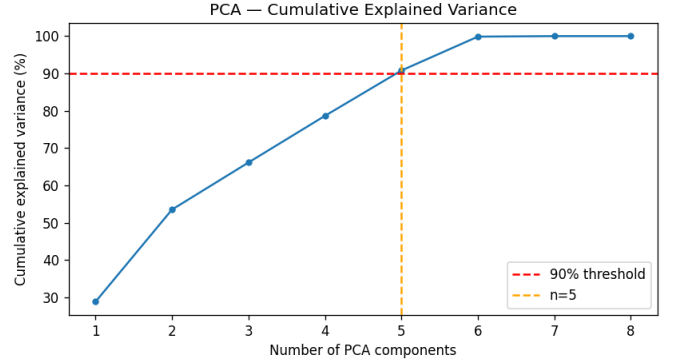


Fig. 1. Cumulative explained variance versus number of PCA components. The red dashed line marks the 90% retention threshold; the orange vertical line indicates the selected $n = 5$ components (90.8% variance retained). The steep initial rise and subsequent plateau confirm that five components provide a compact yet faithful representation of the original 8-D feature space.

B. K-Means

Fig. 2 displays both the inertia (elbow) curve and the Silhouette Score for $k \in \{2, \dots, 12\}$. The inertia curve decreases monotonically and smoothly without a sharp inflection point—a pattern typical of high-dimensional PCA-reduced data where inter-point distances are relatively homogeneous. Because inertia alone is inconclusive in this setting, it is reported for transparency but is not used as the primary selection criterion. The Silhouette Score exhibits a sharp, unambiguous global maximum at $k = 2$ ($\text{Sil} = 0.716$), dropping steeply to 0.304 at $k = 3$ —a reduction of more than 57%—and declining

monotonically to ≈ 0.21 for $k \geq 5$. The magnitude and abruptness of this drop indicate that any partition beyond two clusters introduces artificial fragmentation inconsistent with the underlying density structure.

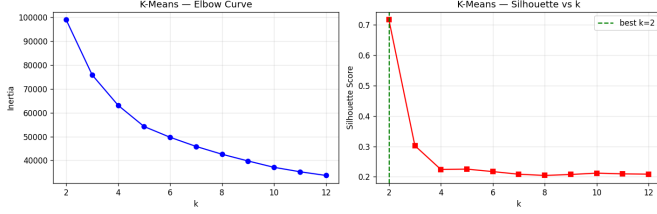


Fig. 2. K-Means hyperparameter selection: inertia elbow curve (left) and Silhouette Score versus k (right) for $k \in \{2, \dots, 12\}$. The inertia curve decreases smoothly without a clear inflection point, while the Silhouette Score peaks sharply at $k = 2$ (green dashed line) and drops by over 57% at $k = 3$, providing unambiguous support for the two-cluster solution.

Stability analysis across seeds $\{42, 123, 777\}$ confirms that the solution is fully deterministic: all three runs produce identical assignments with $\text{Sil} = 0.716 \pm 0.000$, $\text{DB} = 0.397 \pm 0.000$, and $\text{CH} = 5634.4 \pm 0.0$. Zero variance across seeds indicates that the two clusters occupy well-separated, convex regions where k -means++ initialisation consistently locates the global optimum.

The PCA projection in Fig. 3 reveals two geometrically distinct groups. Cluster 0 (dark blue) occupies a broad diagonal band extending from negative to highly positive values along PC1, reflecting the gradual long-run growth in electricity consumption encoded by `year_trend` and `roll12_mean`. Cluster 1 (light blue) is a compact, isolated satellite at ($\text{PC1} \approx -2$, $\text{PC2} \approx 8$ – 12), separated from Cluster 0 by a clear gap of several standard deviations along PC2.

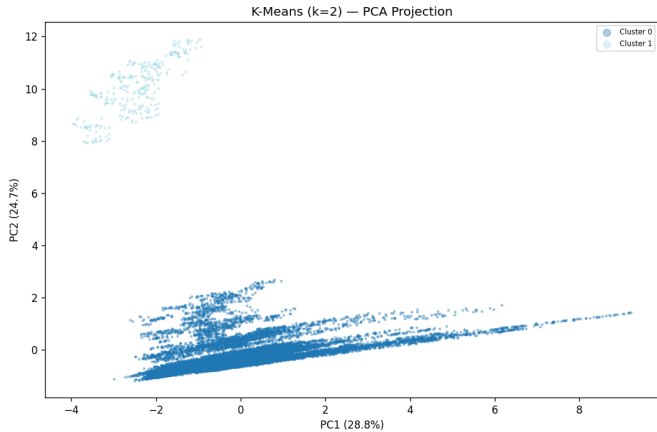


Fig. 3. K-Means ($k = 2$) cluster assignments projected onto PC1 (28.8%) and PC2 (24.7%). Cluster 0 (dark blue) spans a broad diagonal band reflecting the long-run consumption trend; Cluster 1 (light blue) forms a compact, isolated satellite in the upper-left region separated by a clear gap along PC2.

C. DBSCAN

Fig. 4 plots the sorted 5-NN distances for all 17,918 observations in ascending order. The curve remains nearly flat

below 0.2 for the first $\approx 15,000$ points, confirming that the vast majority of observations reside within uniformly dense regions. A pronounced upturn begins near index 16,000, where distances accelerate sharply from ≈ 0.5 to values exceeding 3.5 for the most isolated points. The 90th-percentile elbow heuristic yields $\varepsilon_{\text{elbow}} = 0.547$ (red dashed line), which serves as a lower-bound starting point for the grid search. The grid search selects $\varepsilon = 1.093$ and `min_samples` = 10 as the optimal configuration: a lower ε fragments the elongated Cluster 0 into spurious sub-clusters, while $\varepsilon = 1.093$ preserves a coherent two-cluster solution and correctly isolates the most anomalous observations as noise.

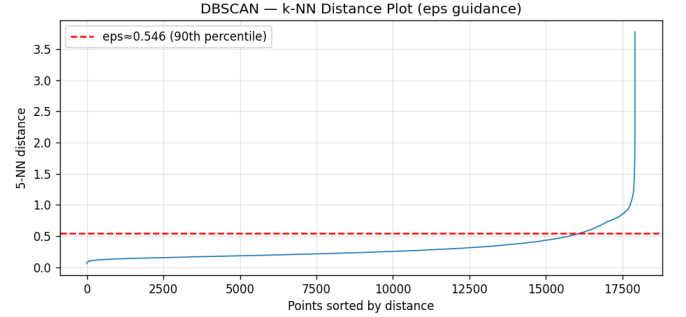


Fig. 4. DBSCAN k -NN distance plot (5-NN distances sorted ascending). The near-flat plateau for the first $\approx 15,000$ points confirms a dominant high-density core; the sharp upturn near index 16,000 marks the boundary between dense-region members and noise candidates. The red dashed line marks the 90th-percentile elbow estimate ($\varepsilon \approx 0.547$); the final $\varepsilon = 1.093$ was selected by composite-score grid search.

The optimal configuration discovers 2 clusters and labels 77 observations (0.43%) as noise. Fig. 5 shows that Cluster 1 (light blue, upper-left satellite) satisfies the core-point criterion and is assigned a proper cluster label, while the 77 noise points (grey) are distributed along the outer boundary of Cluster 0—not within the satellite. This spatial distinction reveals two qualitatively different anomaly types that partition-based methods conflate: the coherent extreme-volatility satellite and sporadic isolated observations that fail to reach the density threshold in any direction. The domain significance of this distinction is discussed in Section VI.

D. Hierarchical Clustering

Fig. 6 shows the truncated Ward-linkage dendrogram computed on a stratified 2,000-observation sample. The most salient feature is the dominant bifurcation at distance ≈ 85 , which joins the two top-level branches into a single root. The next merge below occurs at distance ≈ 70 , leaving a gap of approximately 15 distance units—the largest inter-level gap anywhere in the tree. This gap provides independent, algorithm-agnostic confirmation that $k = 2$ is the natural partition of the data, consistent with the K-Means Silhouette analysis but derived from an entirely different algorithmic principle.

Fitting Agglomerative Clustering with Ward linkage and $k = 2$ on the full 17,918-observation matrix produces a

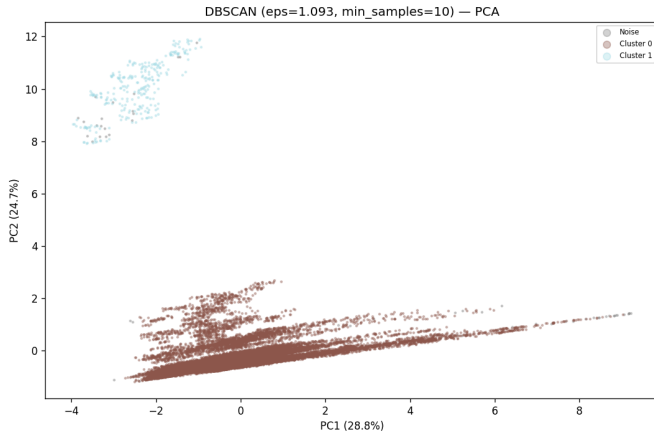


Fig. 5. DBSCAN ($\epsilon = 1.093$, $\text{min_samples} = 10$) assignments projected onto PC1–PC2. Grey points indicate the 77 noise observations (0.43%). Cluster 1 (light blue) occupies the same upper-left satellite region as in K-Means, while noise points are distributed along the outer boundary of Cluster 0 (red-brown), revealing a class of peripheral anomalies invisible to partition-based methods.

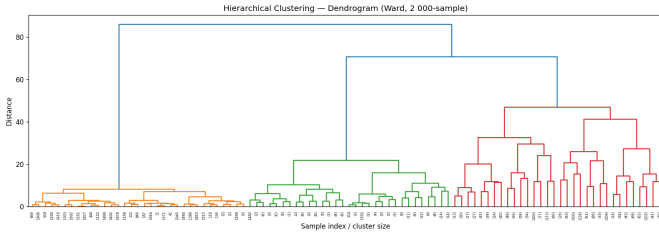


Fig. 6. Truncated Ward-linkage dendrogram (2,000-observation sample). The orange subtree (left) corresponds to Cluster 1; the green and red subtrees together form Cluster 0. The dominant bifurcation at distance ≈ 85 produces the largest inter-level gap in the tree (≈ 15 units), providing algorithm-independent confirmation that $k = 2$ is the natural cut.

partition identical to K-Means ($\text{Sil} = 0.716$, $\text{DB} = 0.397$, $\text{CH} = 5634.4$), as confirmed by the PCA projection in Fig. 7. The agreement between two structurally different algorithms—one centroid-based, the other variance-minimising through bottom-up merging—provides strong evidence that the two-cluster boundary reflects a genuine discontinuity in the data.

E. Quantitative Comparison

Table II summarises the best-configuration metrics for all three algorithms. DBSCAN achieves the highest Silhouette Score (0.721) and the lowest Davies–Bouldin Index (0.390), reflecting marginally tighter cohesion and better separation attributable to the exclusion of the 77 noise points from the cluster interior. The Calinski–Harabasz Index is slightly lower for DBSCAN (5428.3 vs. 5634.4), a mechanical consequence of excluding noise points from the between-cluster dispersion ratio rather than a genuine difference in geometric quality. Across all three metrics the differences between algorithms are small in absolute terms, reinforcing that all three discover the same underlying structure.

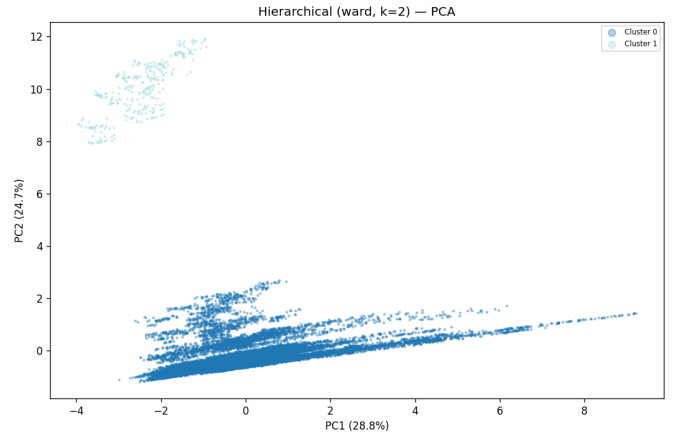


Fig. 7. Hierarchical Clustering (Ward, $k = 2$) PCA projection. The assignment is identical to K-Means (Fig. 3), confirming that the bimodal partition is robust across fundamentally different algorithmic paradigms.

TABLE II
ALGORITHM COMPARISON — BEST CONFIGURATION PER METHOD

Algorithm	k	Noise %	Sil $\uparrow$	DB $\downarrow$	CH $\uparrow$
K-Means	2	0.00%	0.716	0.397	5634.4
DBSCAN	2	0.43%	0.721	0.390	5428.3
Hierarchical	2	0.00%	0.716	0.397	5634.4

F. Feature Ablation

K-Means ($k = 2$) is applied independently to two domain-motivated subsets to isolate the contribution of temporal versus economic signals.

The temporal-subset projection (Fig. 8, top) preserves clean bimodal separation with only four features: Cluster 1 migrates to a distant isolated region at $\text{PC1} > 8$, with a clear empty gap separating it from Cluster 0. PC1 in this projection (33.2% variance) is dominated by the seasonal Fourier encodings, confirming that cyclical seasonal patterns are the primary axis of cluster differentiation.

The economic-subset projection (Fig. 8, bottom) presents a markedly different picture: the two clusters overlap substantially in the central region ($\text{PC1} \in [0, 3]$) with no visible boundary. Although PC1 alone captures 67.5% of variance in this subset—driven by `roll12_mean`—it fails to isolate Cluster 1, which scatters across the full PC1 range. The bimodal structure of U.S. retail electricity data is therefore *primarily seasonal and trend-driven*: temporal features alone are sufficient to recover the partition, while economic features are necessary complements in the full set but cannot reproduce the structure independently.

The ablation results confirm that temporal features are the primary structural driver of the bimodal partition, while economic features provide complementary but insufficient signal in isolation.

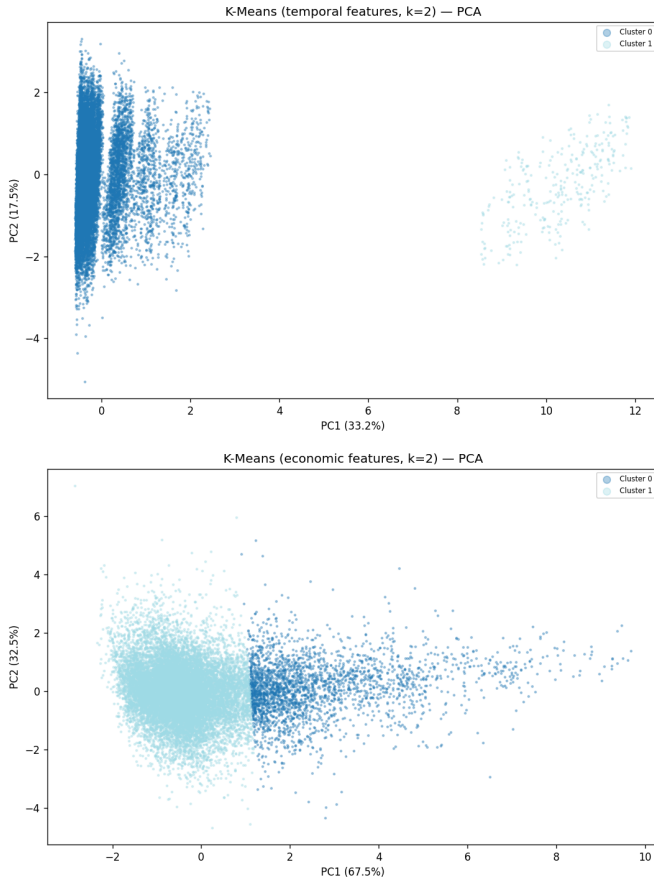


Fig. 8. Feature ablation experiment. *Top*: temporal features only; Cluster 1 is displaced to $PC1 > 8$, preserving clean bimodal separation. *Bottom*: economic features only; substantial cluster overlap confirms that price and revenue features alone cannot recover the partition.

VI. DISCUSSION AND DOMAIN INTERPRETATION

The two clusters carry a coherent domain interpretation. Cluster 0 ($\approx 17,841$ records) corresponds to the *baseline consumption regime*: moderate and stable demand characterised by low rolling volatility, gradual year-over-year growth consistent with long-run electrification trends, and retail prices within the normal historical range. This cluster captures the typical seasonal oscillation of U.S. electricity consumption that is well-documented in the EIA literature.

Cluster 1 captures an *extreme-volatility regime*: state-months with anomalously high `roll12_std`, elevated prices, and consumption spikes that deviate sharply from the 12-month trailing mean. These records are disproportionately concentrated in high-demand summer months in large-consumption states, consistent with the well-documented effect of extreme heat events on load profiles [8]. The compactness and large PC2 separation of this satellite indicate that extreme-volatility episodes form a structurally distinct regime rather than a tail of the normal distribution.

Table III reports the mean value of each feature per cluster, confirming the qualitative interpretation above with concrete numerical evidence.

TABLE III
CLUSTER FEATURE PROFILES — MEAN VALUES (K-MEANS)

Feature	Cluster 0	Cluster 1
	Baseline ($n=17,629$)	Extreme-Volatility ($n=289$)
<code>roll12_mean</code> (MWh)	10,194	310,926
<code>roll12_std</code> (MWh)	1,136	31,129
<code>yoy_growth</code>	0.008	0.008
<code>price</code> (¢/kWh)	10.78	10.14
<code>revenue_per_mwh</code>	0.108	0.101
<code>month_sin</code>	0.002	0.002
<code>month_cos</code>	0.003	0.003
<code>year_trend</code>	157.0	157.0

The 77 DBSCAN noise points (0.43%) represent a third, qualitatively different category. Unlike Cluster 1 members, which satisfy the core-point density criterion, noise points are scattered along the outer boundary of Cluster 0, failing to reach the density threshold in any direction. These peripheral anomalies likely correspond to one-time consumption shocks—such as large industrial facility openings or closings in small-population territories—or to EIA survey reporting revisions. Their identification as noise rather than cluster members is operationally valuable: grid operators and data quality analysts can flag these state-months for manual review without treating them as part of either the baseline distribution or the coherent extreme-volatility regime.

The feature ablation results reinforce this interpretation. The primary axis separating the two clusters reflects cyclical seasonal patterns rather than differences in retail price or revenue per MWh, aligning with the established dominance of weather-driven cooling and heating load in U.S. electricity demand [8].

VII. CONCLUSIONS

A robust bimodal structure governs U.S. retail electricity data. All three algorithms independently converge on $k = 2$: the Silhouette Score peaks sharply at $k = 2$ in the K-Means sweep, the Ward dendrogram exhibits its largest inter-level gap at the two-cluster cut, and DBSCAN discovers exactly two dense regions. The resulting metrics ($Sil \approx 0.72$, $DB < 0.40$, $CH > 5,400$) place this solution firmly in the *well-separated* regime [9], ruling out algorithmic artefact and confirming a genuine discontinuity in the data.

The partition is seasonal in origin. As established by the feature ablation experiment, temporal features alone recover the clean two-cluster separation while economic features produce substantial overlap in isolation. Models that omit seasonal encodings risk conflating the baseline and extreme-volatility regimes characterised in Section VI.

DBSCAN is the recommended algorithm for this domain. It matches K-Means and Hierarchical Clustering quantitatively while uniquely producing the three-way distinction identified in Section VI—baseline regime, extreme-volatility regime, and isolated peripheral anomalies—providing action-

able signal for grid operators and data quality review that partition-based methods cannot deliver.

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